Problem No : 3

Source Code and Output:

```sql
INSERT INTO student (student_id, name, dept)
VALUES
    (1, 'Jiangly', 'CSE'),
    (2, 'tourist', 'EEE'),
    (3, 'Petr', 'CSE'),
    (4, 'Benq', 'CE'),
    (5, 'Errichto', 'EEE'),
    (6, 'Radewoosh', 'CSE'),
    (7, 'Tomek', 'EEE');
```

```sql
    name VARCHAR(200),
    dept VARCHAR(200)
CREATE TABLE enrollments (
    student_id INT,
    course_id INT,
    grade CHAR(3),
    PRIMARY KEY(student_id, course_id),
    FOREIGN KEY(student_id) REFERENCES student(student_id)
);
```

```sql
INSERT INTO enrollments (student_id, course_id, grade)
VALUES
    (1, 101, 'A+'),
    (2, 102, 'A+'),
    (3, 103, 'A'),
    (5, 101, 'A'),
    (3, 101, 'A+');
```

```sql
-- INNER JOIN
SELECT s.student_id, s.name, e.course_id
FROM student AS s
INNER JOIN enrollments AS e
ON s.student_id = e.student_id;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

student_id	course_id	grade
1	101	A+
2	102	A+
3	101	A+
3	103	A
5	101	A
NULL	NULL	NULL

```sql
-- LEFT JOIN
SELECT s.student_id, s.name, e.course_id
FROM student AS s
LEFT JOIN enrollments AS e
ON s.student_id = e.student_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_id	name	course_id
1	Jiangly	101
2	tourist	102
3	Petr	101
3	Petr	103
5	Errichto	101

```sql
-- RIGHT JOIN
SELECT s.student_id, s.name, e.course_id
FROM student AS s
RIGHT JOIN enrollments AS e
ON s.student_id = e.student_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_id	name	course_id
1	Jiangly	101
2	tourist	102
3	Petr	101
3	Petr	103
4	Benq	NULL
5	Errichto	101

```sql
-- OUTER JOIN
SELECT s.student_id, s.name, e.course_id
FROM student AS s
LEFT JOIN enrollments AS e
ON s.student_id = e.student_id
UNION
SELECT s.student_id, s.name, e.course_id
FROM student AS s
RIGHT JOIN enrollments AS e
ON s.student_id = e.student_id;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_id	name	course_id
1	Jiangly	101
2	tourist	102
3	Petr	101
3	Petr	103
5	Errichto	101

```sql
-- Students enrolled in course 101
SELECT s.name, e.course_id
FROM student AS s
INNER JOIN enrollments AS e
ON s.student_id = e.student_id
WHERE e.course_id = 101;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

name	course_id
Jiangly	101
Petr	101
Errichto	101

```sql
-- Students not enrolled in any course
SELECT s.student_id, s.name, e.course_id
FROM student AS s
LEFT JOIN enrollments AS e
ON s.student_id = e.student_id
WHERE e.course_id IS NULL;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

student_id	name	course_id
4	Benq	NULL
6	Radewoosh	NULL
7	Tomek	NULL